

# MYSQL Table Column Level Encryption Steps

2026-03-10

```
mysql> SHOW VARIABLES LIKE 'block_encryption_mode';
+-----+-----+
| Variable_name | Value      |
+-----+-----+
| block_encryption_mode | aes-128-ecb |
+-----+-----+
1 row in set (0.04 sec)
```

## 1. Create database and table

```
CREATE TABLE citizens (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100),
  mobile VARCHAR(15),
  aadhar_no VARCHAR(20)
);
```

```
mysql> create database Encryption_test;
Query OK, 1 row affected (0.01 sec)

mysql> use Encryption_test;
Database changed
mysql> CREATE TABLE citizens (
  ->   id INT AUTO_INCREMENT PRIMARY KEY,
  ->   name VARCHAR(100),
  ->   mobile VARCHAR(15),
  ->   aadhar_no VARCHAR(20)
  -> );
Query OK, 0 rows affected (0.08 sec)

mysql> █
```

Check table

DESC citizens;

## 2. First Insert Data Without Encryption

```
INSERT INTO citizens(name,mobile,aadhar_no)
```

```
VALUES
```

```
('Ajit','9876543210','123412341234'),
```

```
('Ganesh','9123456789','567856785678');
```

Check data

```
mysql> INSERT INTO citizens(name,mobile,aadhar_no)
-> VALUES
-> ('Ajit','9876543210','123412341234'),
-> ('Ganesh','9123456789','567856785678');
Query OK, 2 rows affected (0.03 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

Select \* from citizens;

```
mysql> select * from citizens;
+----+-----+-----+-----+
| id | name  | mobile   | aadhar_no |
+----+-----+-----+-----+
| 1  | Ajit  | 9876543210 | 123412341234 |
| 2  | Ganesh | 9123456789 | 567856785678 |
+----+-----+-----+-----+
2 rows in set (0.01 sec)
```

## 3. Change Column Type for Encryption

Encrypted values must be stored in **VARBINARY**.

```
ALTER TABLE citizens
```

```
MODIFY aadhar_no VARBINARY(255);
```

```
mysql> ALTER TABLE citizens
-> MODIFY aadhar_no VARBINARY(255);
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Now the column can store encrypted data.

⚠ But **existing rows remain unchanged**.

## 4. Create Stored Procedure for Encrypted Insert

```
DELIMITER $$
```

```
CREATE PROCEDURE insert_citizen(
```

```
IN p_name VARCHAR(100),
```

```

IN p_mobile VARCHAR(15),
IN p_aadhar VARCHAR(20)
)
BEGIN
INSERT INTO citizens(name,mobile,aadhar_no)
VALUES(
p_name,
p_mobile,
AES_ENCRYPT(p_aadhar,'Db@d3v2026')
);
END $$
DELIMITER ;

```

```

mysql>
mysql> DELIMITER $$
mysql> CREATE PROCEDURE insert_citizen(
-> IN p_name VARCHAR(100),
-> IN p_mobile VARCHAR(15),
-> IN p_aadhar VARCHAR(20)
-> )
-> BEGIN
-> INSERT INTO citizens(name,mobile,aadhar_no)
-> VALUES(
-> p_name,
-> p_mobile,
-> AES_ENCRYPT(p_aadhar,'Db@d3v2026')
-> );
-> END $$
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>

```

## 5. Insert New Data (Encrypted)

```
CALL insert_citizen('Suraj','9988776655','999988885555');
```

```

mysql>
mysql> CALL insert_citizen('Suraj','9988776655','999988885555');
Query OK, 1 row affected (0.01 sec)

mysql> CALL insert_citizen('Rahul','9988776655','999988887778');
Query OK, 1 row affected (0.01 sec)

```

### 5.a) Read Decrypted Aadhaar Data

DELIMITER \$\$

CREATE PROCEDURE get\_citizens()

BEGIN

SELECT

id,

name,

mobile,

CAST(AES\_DECRYPT(aadhar\_no,'Db@d3v2026') AS CHAR) AS aadhar\_no

FROM citizens;

END \$\$

DELIMITER ;

```
mysql>
mysql> CREATE PROCEDURE get_citizens(
  -> BEGIN
  ->
  -> SELECT
  -> id,
  -> name,
  -> mobile,
  -> CAST(AES_DECRYPT(aadhar_no,'Db@d3v2026') AS CHAR) AS aadhar_no
  -> FROM citizens;
  ->
  -> END $$
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql>
```

Normal select query : in this before encryption we can see the data encrypt

CALL get\_citizens();

As we below see the encrypted column are showing using procedure inside we write decrypt function but the before function the previous two values are showing null because that time there data type are varchar for applying encryption we change it's data type to varbinary that's why it is showing null value but if we only run normal select query that time it will show encrypt data like .

```
mysql> CALL get_citizens();
```

| id | name   | mobile     | aadhar_no    |
|----|--------|------------|--------------|
| 1  | Ajit   | 9876543210 | NULL         |
| 2  | Ganesh | 9123456789 | NULL         |
| 3  | Suraj  | 9988776655 | 999988885555 |
| 4  | Rahul  | 9988776655 | 999988887778 |

```
4 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> select * from citizens;
```

| id | name   | mobile     | aadhar_no                          |
|----|--------|------------|------------------------------------|
| 1  | Ajit   | 9876543210 | 0x313233343132333431323334         |
| 2  | Ganesh | 9123456789 | 0x353637383536373835363738         |
| 3  | Suraj  | 9988776655 | 0xF284F9DB781237A261B71A5E3CD87CD5 |
| 4  | Rahul  | 9988776655 | 0x5ABF3493E41C54088B9F2779462B79FE |

```
4 rows in set (0.00 sec)

mysql>
```

## 6. Check Table Data

```
SELECT id,name,mobile,HEX(aadhar_no)
FROM citizens;
```

```
mysql> SELECT id,name,mobile,HEX( aadhar_no)
-> FROM citizens;
```

| id | name   | mobile     | HEX( aadhar_no)                  |
|----|--------|------------|----------------------------------|
| 1  | Ajit   | 9876543210 | 313233343132333431323334         |
| 2  | Ganesh | 9123456789 | 353637383536373835363738         |
| 3  | Suraj  | 9988776655 | F284F9DB781237A261B71A5E3CD87CD5 |
| 4  | Rahul  | 9988776655 | 5ABF3493E41C54088B9F2779462B79FE |

```
4 rows in set (0.00 sec)
```

## 7. Try Decrypting All Rows

```
SELECT
id,name, CAST(AES_DECRYPT(aadhar_no,'Db@d3v2026') AS CHAR) AS aadhar_no
```

FROM citizens;

## 8. Encrypt Existing Data (Manual Migration)

UPDATE citizens

SET aadhar\_no = AES\_ENCRYPT(aadhar\_no,'Db@d3v2026')

WHERE id IN (1,2);

```
mysql> UPDATE citizens
      -> SET aadhar_no = AES_ENCRYPT(aadhar_no,'Db@d3v2026')
      -> WHERE id IN (1,2);
Query OK, 2 rows affected (0.01 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```

After update the details are change to encrypt and below we have to decrypt to see the data

```
mysql> select * from citizens;
+----+-----+-----+-----+
| id | name  | mobile | aadhar_no |
+----+-----+-----+-----+
| 1  | Ajit  | 9876543210 | 0x1E261B86926F0A57C558EB591D78EAE6 |
| 2  | Ganesh | 9123456789 | 0x12F46476AC86A3A6F449B748BF93E044 |
| 3  | Suraj  | 9988776655 | 0xF284F9DB781237A261B71A5E3CD87CD5 |
| 4  | Rahul  | 9988776655 | 0x5ABF3493E41C54088B9F2779462B79FE |
+----+-----+-----+-----+
4 rows in set (0.01 sec)
```

## 9. Verify Again

SELECT

id, name,

CAST(AES\_DECRYPT(aadhar\_no,'Db@d3v2026') AS CHAR) AS aadhar\_no

FROM citizens;

```
mysql> SELECT id, name, CAST(AES_DECRYPT(aadhar_no,'Db@d3v2026') AS CHAR) AS aadhar_no FROM citizens;
+----+-----+-----+
| id | name  | aadhar_no |
+----+-----+-----+
| 1  | Ajit  | 123412341234 |
| 2  | Ganesh | 567856785678 |
| 3  | Suraj  | 999988885555 |
| 4  | Rahul  | 999988887778 |
+----+-----+-----+
4 rows in set (0.00 sec)
```

## 10. Procedure to See Encrypted Values (HEX) or verify data

DELIMITER \$\$

CREATE PROCEDURE check\_encrypted\_data()

```

BEGIN

SELECT

id,

name,

mobile,

HEX(aadhar_no) AS encrypted_aadhar

FROM citizens;

END $$

DELIMITER ;

```

```

mysql> DELIMITER $$
mysql> CREATE PROCEDURE check_encrypted_data( )
-> BEGIN
-> SELECT
-> id,
-> name,
-> mobile,
-> HEX(aadhar_no) AS encrypted_aadhar
-> FROM citizens;
-> END $$
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>

```

CALL check\_encrypted\_data();

```

mysql> CALL check_encrypted_data();
+-----+-----+-----+-----+
| id | name | mobile | encrypted_aadhar |
+-----+-----+-----+-----+
| 1 | Ajit | 9876543210 | 1E261B86926F0A57C558EB591D78EAE6 |
| 2 | Ganesh | 9123456789 | 12F46476AC86A3A6F449B748BF93E044 |
| 3 | Suraj | 9988776655 | F284F9DB781237A261B71A5E3CD87CD5 |
| 4 | Rahul | 9988776655 | 5ABF3493E41C54088B9F2779462B79FE |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

```

## 11. Procedure to Get One User by ID

```

DELIMITER $$

CREATE PROCEDURE get_citizen_by_id(

IN p_id INT

)

```

```
BEGIN
SELECT
id,
name,
mobile,
CAST(AES_DECRYPT(aadhar_no,'SecureKey') AS CHAR) AS aadhar_no
FROM citizens
WHERE id = p_id;
END $$
DELIMITER ;
CALL get_citizen_by_id(1);
```

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